CSE343 Web Development — Labs 3 & 4 Report

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Submission overview

This report documents the two JavaScript web applications completed from the supplied Lab 3 and Lab 4 task slides. The source code is included in the repository, separated into standalone HTML, CSS, and JavaScript files for each lab.

GitHub repository: https://github.com/Kerolos-eng/cse343-web-development-labs-3-4

Lab 3 — Interactive Calculator Webpage

The Lab 3 task asks for a webpage calculator for basic mathematical operations, implemented using HTML, CSS, and JavaScript. The calculator has a responsive interface and supports addition, subtraction, multiplication, division, decimals, percentage, sign change, clear, and keyboard input. Division by zero is handled safely, and the page updates the display through DOM events.

Screenshot evidence

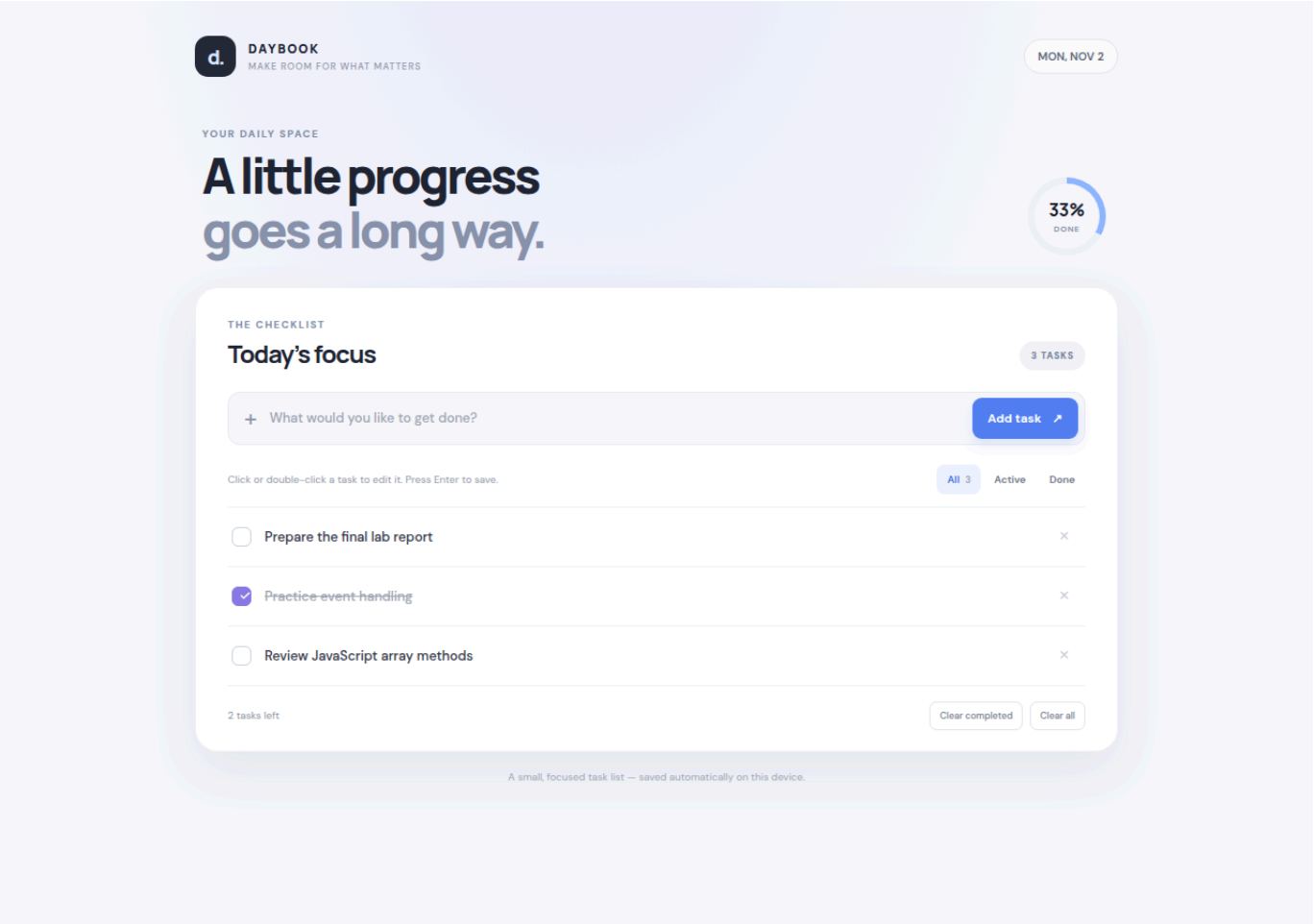


Files: `Lab3/index.html`, `Lab3/lab3-calculator.css`, `Lab3/lab3-calculator.js`

Lab 4 — To-do List Application

The Lab 4 task slide presents a to-do list application. The task manager implements the visible task-list workflow and adds useful interactions: create tasks, mark tasks complete, filter all/active/completed, edit task text, delete individual tasks, clear completed tasks, and clear the whole list. It displays remaining-task and completion counts and persists tasks with `localStorage`.

Screenshot evidence



Files: `Lab4/index.html`, `Lab4/lab4-todo.css`, `Lab4/lab4-todo.js`

Implementation and checks

Both projects are client-side applications and require no server or package installation. They use semantic HTML, responsive CSS, vanilla JavaScript, DOM selection and updates, event listeners, and clear separation of structure, presentation, and behavior. The Lab 4 implementation also uses a `TaskList` class, array `map` / `filter` methods, object spread, template literals, and browser storage.

The JavaScript source files were checked with Node.js syntax validation. The interactive controls and page rendering were checked in a browser before capturing the screenshots.

AI assistance

Manus was used to help create and revise the application code and styling, prepare this report, and run browser-based checks. The student is responsible for reviewing the work,

understanding it, and following the course policy on AI assistance before submission.

Included files and final check

The repository and submission bundle include both applications' source files, screenshot evidence, this report, and local run instructions in the README. The report includes the supplied student name and ID. Confirm that your instructor can access the private repository if a repository link is required.